

# DoNOF Input Reference

This document summarizes the current DoNOF input namelists and is intended as a practical reference for users preparing input files.

## Overview

DoNOF input is organized around three namelists:

- **&INPRUN**: run type, integral engine, ERI/RI options, electric field, and general limits
- **&NOFINP**: NOF settings, orbital optimization, perturbative options, restart behavior, and output controls
- **&INPDYN**: molecular-dynamics controls, used only when **RUNTYP='DYN'**

Most jobs use **&INPRUN**, **\$DATA**, and **&NOFINP**. Dynamics jobs also use **&INPDYN**.

## Minimal Examples

### Single-point energy

```
&INPRUN RUNTYP='ENERGY' MULT=1 ICHARG=0 /
$DATA
Water
aug-cc-pVDZ
O 8.0  0.000000  0.000000  0.000000
H 1.0  0.000000  0.757000  0.587000
H 1.0  0.000000 -0.757000  0.587000
$END
&NOFINP IPNOF=8 /
```

### Geometry optimization

```
&INPRUN RUNTYP='OPTGEO' /
$DATA
Water optimization
aug-cc-pVDZ
O 8.0  0.000000  0.000000  0.000000
H 1.0  0.000000  0.757000  0.587000
H 1.0  0.000000 -0.757000  0.587000
$END
&NOFINP IPNOF=8 MOLDENGEO=1 /
```

### Transition-state optimization

```
&INPRUN RUNTYP='TSOPT' /
$DATA
Transition state guess
aug-cc-pVDZ
...
$END
&NOFINP IPNOF=8 MOLDENGEO=1 /
```

## Molecular dynamics

```
&INPRUN RUNTYP='DYN' MULT=1 ICHARG=0 /
$DATA
Dynamics example
cc-pVDZ
F 9.0 -0.3000 -6.0000 0.0000
H 1.0 -0.3707 0.0000 0.0000
H 1.0 0.3707 0.0000 0.0000
$END
&NOFINP IPNOF=8 Imod=1 /
&INPDYN dt=0.1 tmax=50.0 Vxyz=0,0.1,0,0.025,0,0,-0.025 /
```

Dynamics runs require a valid GCF file to initialize occupations and orbitals.

## ERPA calculation

```
&INPRUN RUNTYP='ENERGY' /
$DATA
ERPA example
aug-cc-pVDZ
...
$END
&NOFINP IPNOF=8 ERPA=.TRUE. /
```

## Nonlinear optical properties

```
&INPRUN RUNTYP='ENERGY' NLOP=1 NPOINT=9 STEP=1.0d-4 /
$DATA
NLOP example
aug-cc-pVDZ
...
$END
&NOFINP IPNOF=8 /
```

## ‘&INPRUN‘

### Main job controls

- RUNTYP='ENERGY'
  - ENERGY: single-point energy
  - GRAD: analytic nuclear gradient
  - OPTGEO: geometry optimization
  - HESS: numerical Hessian from analytic gradients
  - TSOPT: first-order saddle-point search
  - DYN: Born-Oppenheimer on-the-fly molecular dynamics
- MULT=1: spin multiplicity
- ICHARG=0: molecular charge
- IECP=0
  - 0: all-electron calculation
  - 1: read ECP from \$ECP
- IEMOM=1
  - 1: dipole moment

- 2: dipole and quadrupole
- 3: dipole, quadrupole, and octopole

Current defaults include `ERITYP='RI'`, `USELIB=.TRUE.`, `GTYP='SPH'`, and `IPNOF=8`.

## Nonlinear optical properties

- `NLOP=0`
  - -1: compute `alpha`, `beta`, and `gamma`
  - 0: disabled
  - 1: polarizability `alpha`
  - 2: first hyperpolarizability `beta`
  - 3: second hyperpolarizability `gamma`
- `NPOINT=9`: number of field values used in the dyadic Romberg-Richardson procedure
- `STEP=1.0d-4`: initial electric-field step
- `ISOALPHA=0`: request isotropic and anisotropic polarizability analysis

`NLOP /= 0` is only valid with `RUNTYP='ENERGY'`.

## Coordinates and electric field

- `UNITS='ANGS'`
  - `ANGS`
  - `BOHR`
- `EVEC=0.0d0,0.0d0,0.0d0`: electric field in a.u.

## Integral engine and basis conventions

- `USELIB=.TRUE.`
  - `.TRUE.:` `LIBCINT`
  - `.FALSE.:` `HONDO`
- `GTYP='SPH'`
  - `SPH`: spherical Gaussians, valid only with `USELIB=.TRUE.`
  - `CART`: Cartesian Gaussians
- `USEHUB=.FALSE.:` Hubbard-model mode

## ERI and RI controls

- `DONTW=.TRUE.:` do not store two-electron integrals on disk
- `ERITYP='RI'`
  - `FULL`: four-center ERIs
  - `RI`: RI approximation
  - `MIX`: RI first, then full ERIs after convergence
- `CUTOFF=1.0d-9`: Schwarz screening threshold
- `RITYP='JKFIT'`
  - `JKFIT`
  - `GEN`
  - `RIFIT`
- `GEN='A2*':` generative auxiliary basis if `RITYP='GEN'`
- `SMCD=.FALSE.:` symmetric modified Cholesky decomposition for the RI `G` matrix

## Hessian and thermochemistry

- HSSCAL=.TRUE.: in OPTGEO, compute a Hessian from analytic gradients and perform vibrational analysis
- PROJECT=.TRUE.: project out rotational and translational contaminants from the Hessian
- ISIGMA=1: rotational symmetry number

## Allocation limits

- NATmax=100
- NSHELLmax=500
- NPRIMImax=2000
- NSHELLAUXmax=1000
- NPRIMIAUXmax=1000

## ‘&NOFINP‘

## Global optimization

- MAXIT=1000: maximum OCC-SCF iterations
- MAXIT21=3: initial NO-only stage when ICOEF=21
- ICOEF=1
  - 0: optimize occupations only
  - 1: optimize occupations and orbitals
  - 2: optimize orbitals only
  - 21: first run a short orbital-only stage, then switch to ICOEF=1
  - 3: fragment-oriented mode
- ISOFTMAX=1
  - 0: trigonometric occupation parametrization
  - 1: softmax occupation parametrization
- IORBOPT=2
  - 1: iterative diagonalization
  - 2: ADAM
  - 3: AdaBelief
  - 4: YOGI
  - 5: DEMON
  - 6: SQP
- IEINI=0: stop after the initial energy evaluation
- N01=0
  - -1: include core orbitals as fully occupied
  - 0: all orbitals considered
  - positive integer: user-defined number of fully occupied orbitals

## Hartree-Fock guess

- IRHF=1
  - 0: do not obtain HF orbitals
  - 1: HF SCF
  - 2: ADAM-based orbital optimization
  - 3: iterative diagonalization
- NCONVRHF=5

- MAXITRHF=100
- HFDAMP=.TRUE.
- HFEXTRAP=.TRUE.
- HFDIIS=.TRUE.
- KOOPMANS=0

IRHF=1 is not allowed with a nonzero electric field. In OPTGEO, HESS, and DYN, IRHF may be forced to 0.

### Functional selection

- IPNOF=8
  - 5: PNOF5
  - 6: PNOF6
  - 7: PNOF7
  - 8: GNOF
- Ista=0
  - 0: PNOF7
  - 1: PNOF7s
- Imod=0
  - 0: GNOF
  - 1: GNOFm
- HighSpin=.FALSE.
- NCWO=-1
  - 1,2,3,...
  - -1: automatic NVIR/NDOC

For IPNOF=8, only Imod=0 and Imod=1 are accepted.

### Convergence thresholds

- NTHRESHL=4
- NTHRESHE=8
- NTHRESHEC=8
- NTHRESHEN=8

These correspond to:

- THRESHL = 10 raised to (-NTHRESHL)
- THRESHE = 10 raised to (-NTHRESHE)
- THRESHEC = 10 raised to (-NTHRESHEC)
- THRESHEN = 10 raised to (-NTHRESHEN)

Notes:

- in DYN, if NTHRESHE is left at the default 8, it is increased to 10
- NTHRESHEC < 8 is promoted to 8

### Iterative-diagonalization controls

- MAXLOOP=10
- SCALING=.TRUE.
- AUTOZEROS=.TRUE.
- NZEROS=0

- NZEROSm=5
- NZEROSr=2
- ITZITER=10
- DIIS=.TRUE.
- NTHDIIS=3
- NDIIS=5
- PERDIIS=.TRUE.
- DAMPING=.FALSE.
- EXTRAP=.FALSE.

### **ADAM-family controls**

- LR=0.01d0
- FACT=0.2d0
- BETA1=0.7d0
- BETA2=0.9d0

### **Perturbative and response options**

- ERPA=.FALSE.
- OIMP2=.FALSE.
- MBPT=.FALSE.
- NO1PT2=-1
- SC2MCPT=.FALSE.
- NEX=0

Notes:

- APSG and SC2MCPT are only valid for IPNOF=5
- OIMP2 is disabled automatically in DYN

### **Restart and initial guesses**

- RESTART=.FALSE.
- INPUTGAMMA=0
- INPUTC=0
- INPUTFMIUG=0
- INPUTCXYZ=0

When RESTART=.TRUE., the code forces:

- INPUTGAMMA=1
- INPUTC=1
- INPUTFMIUG=1
- INPUTCXYZ=1

In DYN:

- the code forces RESTART=.FALSE.
- INPUTGAMMA=1
- INPUTC=1
- INPUTFMIUG=1 only when IORBOPT=1
- INPUTCXYZ=0

A valid GCF file is still required to initialize a dynamics run.

## Output controls

- NPRINT=0
- IWRITEC=0
- IMULPOP=0
- PRINTLAG=.FALSE.
- DIAGLAG=.FALSE.
- IEKT=0
- IAIMPAC=0
- IFCHK=1
- MOLDEN=1
- MOLDENGEO=0
- INICOND=1
- NOUTRDM=0
- NTHRESHDM=6
- NSQT=1
- NOUTCJK=0
- NTHRESHCJK=6
- NOUTTijab=0
- NTHRESHTijab=6
- APSG=.FALSE.
- NTHAPSG=10

MOLDENGEO=1 saves one Molden file per geometry visited in OPTGEO or TSOPT as `snapshot-####.mld`. It also forces MOLDEN=1.

In DYN:

- IAIMPAC=0
- MOLDEN=0
- MOLDENGEO=0
- INICOND=0
- IFCHK=0

## Miscellaneous controls

- ORTHO=.TRUE.
- CHKORTHO=.FALSE.
- FROZEN=.FALSE.
- IFROZEN=0
- ICGMETHOD=1
  - 1: SUMSL / CG
  - 2: NAG-based path, available only if explicitly enabled
  - 3: LBFGS

## Excited-state ensemble controls

- NEST=0
- OMEGA1=1.0d0

## ‘&INPDYN’

This namelist is used only when RUNTYP=’DYN’.

- `velflag='F'`: constant-velocity dynamics
- `dt=1.0d0`: time step in fs
- `tmax=100.0d0`: total propagation time in fs
- `Vxyz=0.0d0`: initial Cartesian velocities per atom
- `resflag='F'`: restart molecular dynamics
- `snapshot='F'`: save `snapshot-<time>.mld`
- `jumptol=1.0d-4`: threshold for treating a potential-energy jump as a real event
- `energybound='T'`: enable velocity rescaling and event guard
- `integrator='BEEVER'`
  - BEEVER
  - VERLET

## Notes

- `&INPDYN` is read only when `RUNTYP='DYN'`
- `resflag='T'` reads the last geometry and velocities from `DYN1.xyz`
- `snapshot='T'` writes Molden snapshots during the trajectory
- `energybound` accepts only T or F
- `integrator` accepts only BEEVER or VERLET